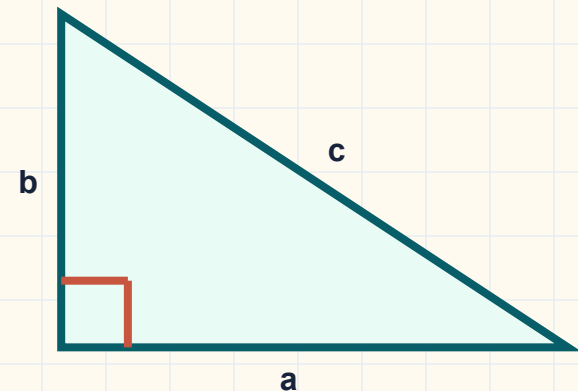


Unlocking the Pythagorean Theorem

Slide 1/10

Hook: How can a builder find a slanted distance without measuring the slant directly?

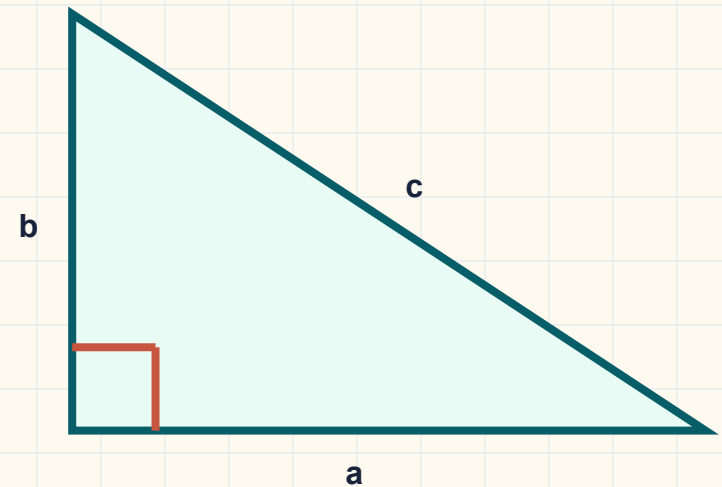
Purpose: connect the lesson to real construction and diagonal-distance situations.



What Makes a Right Triangle?

Slide 2/10

- Right angle: the square corner.
- Legs: the two sides that meet at the right angle.
- Hypotenuse: the side opposite the right angle.



Quick Review: Squares and Square Roots

Slide 3/10

$$3^2 = 9$$

$$4^2 = 16$$

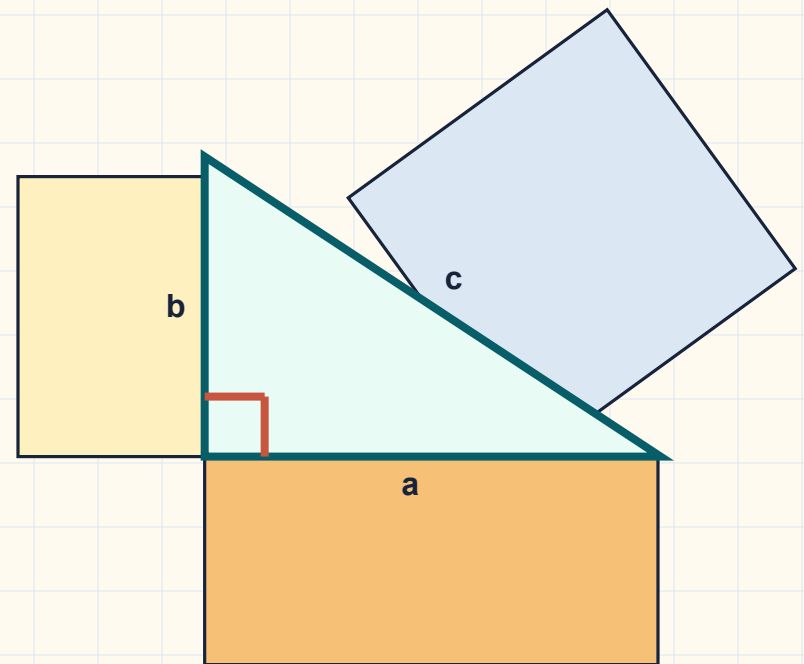
$$\text{sqrt}(25) = 5$$

Squaring finds square area. Square roots move back to side length.

The Area Idea

Slide 4/10

- The square on side a plus the square on side b matches the square on side c.
- This is an area-of-squares relationship.

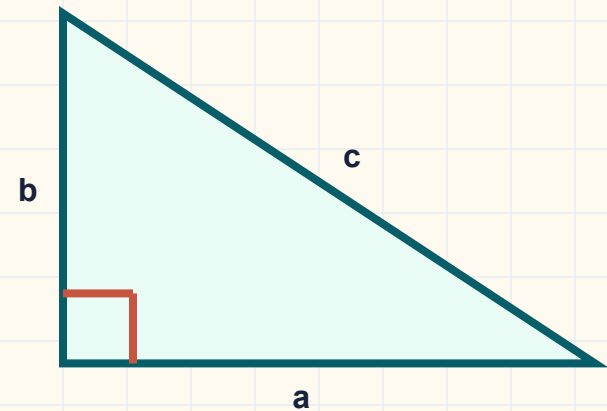


The Formula

Slide 5/10

$$a^2 + b^2 = c^2$$

- a and b are the legs.
- c is the hypotenuse, opposite the right angle.



Worked Example: 3-4-5 Triangle

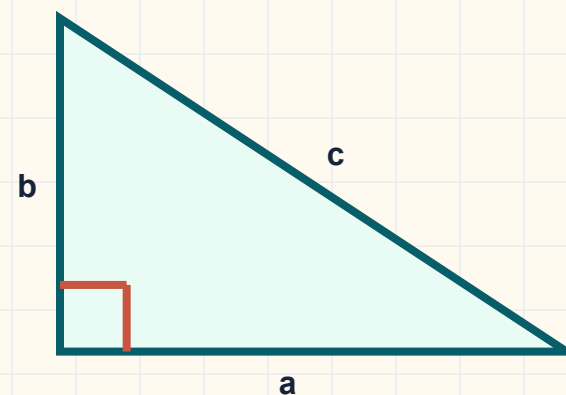
Slide 6/10

$$3^2 + 4^2 = c^2$$

$$9 + 16 = c^2$$

$$25 = c^2$$

$$c = 5$$



Check a Right Triangle: 6-8-10

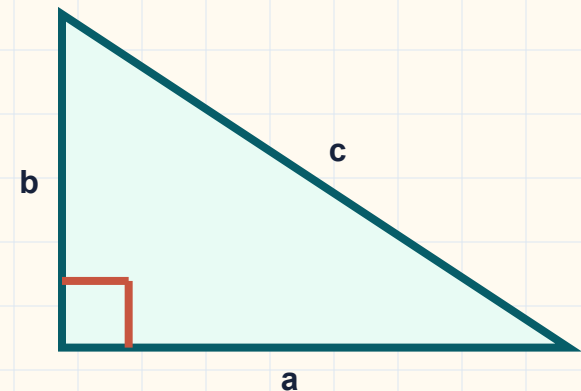
Slide 7/10

$$6^2 + 8^2 = 10^2$$

$$36 + 64 = 100$$

$$100 = 100$$

Therefore, it is a right triangle.



Your Turn: 5-12-13

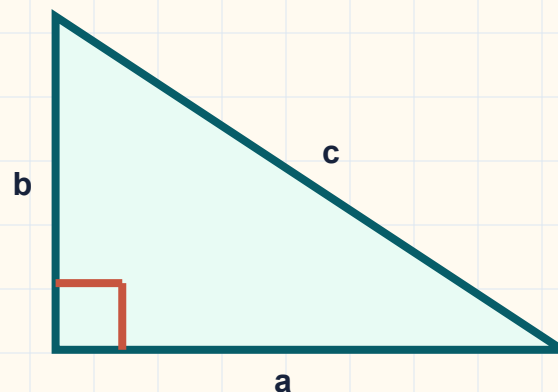
Slide 8/10

Pause and solve for c using legs 5 and 12.

$$5^2 + 12^2 = c^2$$

$$\underline{\quad} + \underline{\quad} = c^2$$

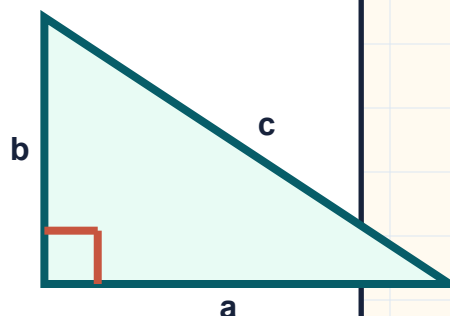
$$c = \underline{\quad}$$



Guided Notes Handout - Part 1

Slide 9/10

- Label a, b, and c.
- Write the formula.
- Copy the worked 3-4-5 example.



- Try the student practice problem.
- Write one question to bring to class.
- Class time will be used for the escape room and teacher support.

Bring one question. In class, your group unlocks the escape room.